1. **Aim/Overview of the practical**: Create the university application to represent course structure to the students

2. **Objective of the practical:** To provide students with a clear and structured view of departments, courses, and modules for effective learning and academic planning.

3. **Code for experiment/practical:**

MODULE SCHEMA

```
 const mongoose = require('mongoose');
const Schema = mongoose.Schema;

const ModuleSchema = new Schema({
  title: { type: String, required: true, trim: true },
  code: { type: String },
  content: { type: String },
  credits: { type: Number, default: 0 },
  course: { type: Schema.Types.ObjectId, ref: 'Course', required: true },
  createdAt: { type: Date, default: Date.now }
});

module.exports = mongoose.model('Module', ModuleSchema);
```

Department Schema

```
const mongoose = require('mongoose');
const Schema = mongoose.Schema;

const DepartmentSchema = new Schema({
  name: { type: String, required: true, unique: true, trim: true },
  description: { type: String },
  createdAt: { type: Date, default: Date.now }
});

module.exports = mongoose.model('Department', DepartmentSchema);
```

Courses Schema

```
const mongoose = require('mongoose');
const Schema = mongoose.Schema;

const CourseSchema = new Schema({
  title: { type: String, required: true, trim: true },
  code: { type: String, required: true, trim: true },
  description: { type: String },
  department: { type: Schema.Types.ObjectId, ref: 'Department', required: true },
  createdAt: { type: Date, default: Date.now }
});

module.exports = mongoose.model('Course', CourseSchema);
```

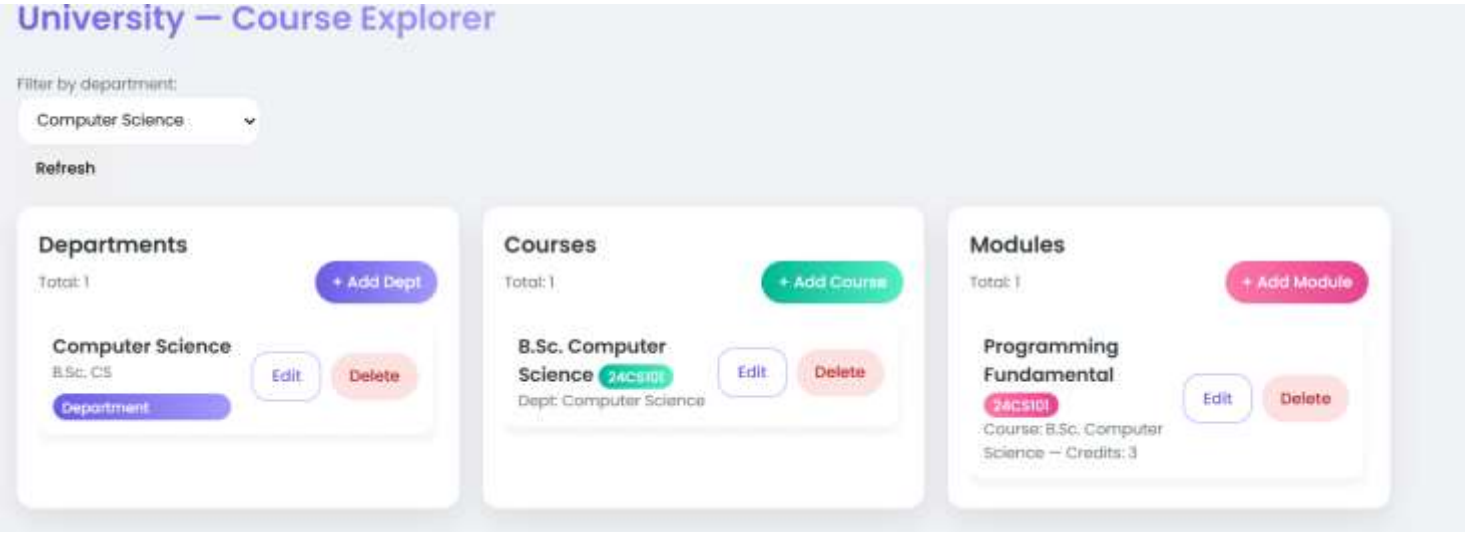
Server.js

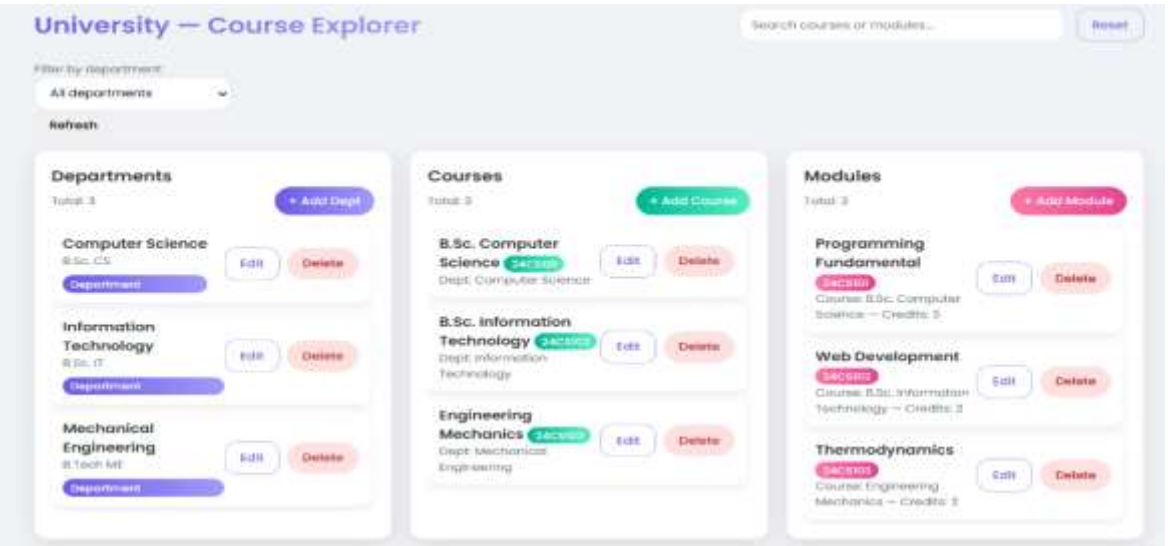
```
require('dotenv').config();
const app = require('./app');
const connectDB = require('./config/database');

const PORT = process.env.PORT || 5000;
const MONGO_URI = process.env.MONGO_URI ||
'mongodb://localhost:27017/university_course_app';

connectDB(MONGO_URI).then(() => {
  app.listen(PORT, () => console.log(`Server running on port ${PORT}`));
});
```

4. Output:





5. Learning Outcomes (What I have learnt):

1. Students will be able to explore different departments and their offered courses easily.
2. They will understand the structure of each course, including its core and elective modules.
3. The app will help students identify learning paths and select relevant subjects.
4. Learners will gain better clarity in academic progression and career preparation.